



JS FOUNDATIONAL HANDBOOK : By Gagan Baghel

Topics —

- ✓ Variables
- ✓ Identifiers
- ✓ Operators
- ✓ Conditional Statements (if/else)
- ✓ Loops
- ✓ null & undefined

✓ SECTION 1 — Concepts (Short Notes)

1. Variables

A variable is a container for storing a value.

Allowed declarations:

```
let age = 21;  
const PI = 3.14;  
var score = 50; // avoid but okay for beginners
```

2. Identifiers

Rules for naming variables:

- Must start with **letter**, **_**, **\$**
- Cannot start with a number
- Cannot contain spaces

- Cannot use reserved keywords
- Must be meaningful

3. Operators

Arithmetic:

`+ - * / % **`

Assignment:

`=, +=, -=, *=, /=, %=`

Comparison:

`==, ===, !=, !==, >, <, >=, <=`

Logical:

`&& || !`

4. Conditional Statements

```
if(condition) { }  
else if(condition) { }  
else { }
```

5. Loops

```
for  
while  
do-while
```

6. null vs undefined

undefined:

Variable declared but not assigned.

null:

Manually empty value.

SECTION 2 — 60 QUESTIONS (Perfect Leveling)

All questions depend only on the above topics.

A. VARIABLE & IDENTIFIER (Beginner)

Q1: Create 5 valid variables and assign numeric values.

Q2: Create 5 invalid identifiers and rewrite them correctly.

Q3: What is wrong in this?

```
let 9age = 20;
```

Q4: Declare a variable using `let` without assigning any value. Log it.

Q5: Create two variables:

```
height = 180  
weight = 70
```

Print both.

Q6: Try to reassign a `const` variable and observe error.

Q7: Which variable names are valid?

- a) total-amount
- b) _age
- c) \$price
- d) let

Q8: Store your age in `let age` and then update it with:

```
age += 1;
```

Q9: Why does this fail?

```
let name;  
let name;
```

Q10: Declare a variable and assign null.

★ B. ARITHMETIC & ASSIGNMENT OPERATORS

Q11: Take two numbers and find:

```
sum  
difference  
product  
division  
remainder
```

Q12: Use `+=` to increase a number by 10.

Q13: Use `=` to decrease a number by 5.

Q14: Multiply a number by 3 using `=`.

Q15: Divide a number by 2 using `/=` .

Q16: What is:

```
10 % 3
```

Q17: Predict:

```
6 ** 2
```

Q18: If:

```
let x = 5;  
x += 5;
```

What is x?

Q19: If:

```
let a = 10;  
a /= 2;
```

Q20: Write code to swap:

```
let a = 5  
let b = 7
```

(without using arrays or temp variables NOT allowed → must use 3rd variable)



C. COMPARISON & LOGICAL OPERATORS

Q21: Compare:

```
5 == "5"  
5 === "5"
```

Q22: Compare:

```
10 != 8  
10 !== 8
```

Q23: Check which number is bigger using `>`.

Q24: Check if a number is positive, negative, or zero using if-else.

Q25: Check if age is between 18 and 25 using `&&`.

Q26: If `marks >= 33` print "Pass", else "Fail".

Q27: Use OR operator:

```
if (isWeekend || isHoliday)
```

Q28: Use NOT operator:

```
let isRaining = false;  
console.log(!isRaining)
```

Q29: Predict:

```
10 > 5 && 2 > 3
```

Q30: Predict:

```
5 == 5 || 6 == 10
```

★ D. CONDITIONAL STATEMENTS ONLY

Q31: If number > 50 print "High".

Q32: If number < 10 print "Low".

Q33: If two numbers are equal print "Equal".

Q34: Check if number is even or odd using `%`.

Q35: Check if a number is divisible by 5.

Q36: Given:

```
let speed = 80;
```

If speed > 60 print "Overspeed".

Q37: Nested if:

Check:

- If number is > 0
- If number is even

Q38: Compare three numbers and print the largest.

Q39: Check if a number is between 1 and 100.

Q40: Check discount eligibility:

```
price >= 1000
```

★ E. LOOPS (Very Beginner-Friendly)

Q41: Print numbers from 1 to 10 using `for`.

Q42: Print numbers from 10 to 1.

Q43: Print even numbers from 1 to 20.

Q44: Print odd numbers from 1 to 20.

Q45: Print numbers from 1 to n (n stored in variable).

Q46: Print sum of first 10 natural numbers.

Q47: Print multiplication table of 5.

Q48: Use `while` loop to print:

```
1 2 3 4 5
```

Q49: Use `do-while` to print numbers from 1 to 5.

Q50: Print all multiples of 3 from 1 to 30.

★ F. null & undefined ONLY

Q51: What is the value of a variable declared but not assigned?

Q52: Create:

```
let a;  
console.log(a);
```

Q53: Set:

```
let x = null;  
console.log(x);
```

Q54: Predict:

```
console.log(undefined + 10);
```

Q55: Predict:

```
console.log(null + 10);
```

Q56: What is:

```
typeof undefined
```

Q57: What is:

```
typeof null
```

Q58: Difference:

undefined vs not declared variable

Q59: Check:

```
if (value == undefined)
```

Q60: Why does `undefined + undefined` **result in NaN?**